CS532 Quiz 4: Feature Tracking

Name: _____

Question 1. You're tracking features across a long monocular video with exposure auto-adjustment, which causes frame-to-frame brightness shifts.

1. Propose modifications to the basic brightness-constancy formulation that improve robustness here and argue how it addresses violations of brightness constancy without breaking the linearization used by KLT

   Solution:

   The usual KLT assumption of brightness constancy is $I_2(x+u) \approx I_1(x)$ breaks when using auto-exposure settings. This is because auto-exposure rescales and shifts the image intensities.

   One way to get around this is to normalize each patch before applying the KLT algorithm:

   $$\tilde{I}(x) = \frac{I(x) - \mu}{\sigma}$$

   where μ and σ is the mean and std. deviation of a patch. After this modification, the usual brightness constancy holds: $\tilde{\mathbf{I}}_\mathbf{2}(\mathbf{x}+\mathbf{u}) = \tilde{\mathbf{I}}_\mathbf{1}(\mathbf{x})$